Positional Mechanisms in Rotary Attention: two factors, one scan

Notes derived from the implementation

Abstract

Rotary attention writes every logit as a sum of products $\mu \cos \varphi$. **A positional mechanism is a choice about one of those two factors, and nothing else.** MapFormer's path integration, Selective RoPE and ordinary RoPE all set the *phase* φ; PoPE sets the *magnitude* μ; MapPoPE is their conjunction and not a third mechanism.

The phase mechanisms are one machine. Each maintains a state by cumulative summation and reads it out linearly into the phase, $s_t = s_{t-1} + \varphi(x_{t-K+1:t})$, $\theta_t = As_t$, and they differ only in the *state rank* m, the nonlinearity of the increment, and its temporal support K. RoPE, MapFormer and Selective RoPE are then *nested*, each inclusion removing a constraint rather than adding a mechanism, so every comparison between them asks whether a constraint is worth keeping and none of them is about expressivity. For MapFormer the reduction is an identity: all 64 of its phase channels are linear readouts of a single *two*-dimensional accumulator.

Rank is the axis that binds, and it binds twice. Confining the phase to an r-dimensional subtorus forces the N^D positions of a D-torus apart by no more than $\sim N^{-\max(D/r,1)}$, so *geometry* sets a hard threshold at $r = D$ and says nothing above it; the measured cost of $r = 2$ in two dimensions is therefore not geometric but an *optimisation* failure, and the two are separable exactly this way. §2.5 states both, with the correction that the first version of the geometric argument — an appeal to a kernel — was wrong and was falsified by its own gate.

The asymmetry between the two factors matters. MapFormer adds a term to the phase and leaves the rest of attention untouched. PoPE replaces the magnitude *and* eliminates the content-dependent intrinsic phase that ordinary rotary attention carries, so it is a *restriction* on the object MapFormer rotates rather than a parallel choice — a plausible source of the sub-additivity seen on text (§4).

Formulas are transcribed from the implementation. Where it departs from a published equation or from canonical RoPE, that is stated at the point of departure. Empirical claims carry their seed counts and minimum detectable effects; several are recorded here only to be marked unsupported.

1 The shared frame

Let $x_1, \dots, x_T \in \mathbb{R}^d$ be token embeddings, and for a single head write queries and keys $q_t = W_Q x_t$, $k_s = W_K x_s \in \mathbb{R}^{d_{\text{head}}}$. Group the d_{head} coordinates into n_b two-dimensional blocks and identify each block with a complex number,

$$\tilde{q}_t^{(c)} \;=\; q_{t,2c} + i\, q_{t,2c+1}, \qquad \tilde{k}_s^{(c)} \;=\; k_{s,2c} + i\, k_{s,2c+1}, \qquad c = 0, \dots, n_b - 1. \tag{1}$$

A rotary scheme attaches a per-position angle $\theta_t^{(c)}$ and rotates:

$$\begin{pmatrix} q'_{t,2c} \\ q'_{t,2c+1} \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\theta_t^{(c)} & -\sin\theta_t^{(c)} \\ \sin\theta_t^{(c)} & \cos\theta_t^{(c)} \end{pmatrix}}_{R(\theta_t^{(c)}) \in SO(2)} \begin{pmatrix} q_{t,2c} \\ q_{t,2c+1} \end{pmatrix}, \tag{2}$$

and likewise for k_s. The unnormalised attention logit is then

$$a_{ts} \;=\; \frac{1}{\sqrt{d_{\text{head}}}}\,\langle q'_t,\, k'_s\rangle \;=\; \frac{1}{\sqrt{d_{\text{head}}}}\sum_{c=0}^{n_b-1}\big\langle R(\theta_t^{(c)})\,q_t^{(c)},\; R(\theta_s^{(c)})\,k_s^{(c)}\big\rangle. \tag{3}$$

Because R is orthogonal, $R(\alpha)^\top R(\beta) = R(\beta-\alpha)$, so

$$a_{ts} \;=\; \frac{1}{\sqrt{d_{\text{head}}}}\sum_c \Big\langle q_t^{(c)},\, R\Big(\theta_s^{(c)}-\theta_t^{(c)}\Big)\,k_s^{(c)}\Big\rangle \;=\; \frac{1}{\sqrt{d_{\text{head}}}}\sum_c \operatorname{Re}\Big[\overline{\tilde q_t^{(c)}}\;\tilde k_s^{(c)}\;e^{i(\theta_s^{(c)}-\theta_t^{(c)})}\Big]. \tag{4}$$

Only the *difference* of angles survives: this is the relative-position property, and it holds for *any* choice of θ.

Finally, write each complex factor in polar form, $\tilde q_t^{(c)} = \mu^q_{t,c}\,e^{i\psi^q_{t,c}}$ and similarly for k. Then (4) becomes

$$\boxed{a_{ts} \;=\; \frac{1}{\sqrt{d_{\text{head}}}}\sum_c \underbrace{\mu^q_{t,c}\,\mu^k_{s,c}}_{\text{magnitude}}\;\cos\big(\underbrace{\theta_s^{(c)}-\theta_t^{(c)}+\psi^k_{s,c}-\psi^q_{t,c}}_{\text{phase}}\big).} \tag{5}$$

Equation (5) is the frame for everything that follows. **Each mechanism is a decision about where the magnitude comes from and where the phase comes from.**

2 The phase axis: a family of scans

Every mechanism that sets θ in (5) does so the same way. Call a **phase generator** the triple (m, φ, K) producing

$$\boxed{s_t \;=\; s_{t-1} + \varphi(x_{t-K+1},\dots,x_t) \;\in\; \mathbb{R}^m, \qquad \theta_t \;=\; A\,s_t \;\in\; \mathbb{R}^{Hn_b}.} \tag{6}$$

m is the **state rank**, φ the increment map, K its temporal support and A a fixed linear readout. Cumulative summation is linear, so it commutes with any linear map: the accumulation may always be pushed down to the smallest space the increment factors through, and m is that dimension. Three named mechanisms are instances, and §2.1–2.3 give them in turn.

generator	m	φ	K	A
RoPE	1	$\equiv 1$	1	ω (frozen)
MapFormer	r	$W_{\text{in}}x$	1	$\operatorname{diag}(\omega)\,W_{\text{out}}$
Selective RoPE	Hn_b	$\sigma(W_g x)\odot(\kappa * W_\omega x)$	$4^\dagger$	τI

† our choice, not theirs: the paper states no convolution width, and its own derivation implies $K=2$. See §2.6.

Because the state is accumulated and the readout is linear, all three are $O(\log T)$ parallel scans. That property is shared and is not what distinguishes them.

2.1 RoPE: the fixed clock

Standard RoPE fixes the angle to be linear in the token index,

$$\theta_t^{(c)} \;=\; t\cdot\omega_c, \qquad \omega_c \;=\; \text{base}^{-2c/d_{\text{head}}} \;=\; \text{base}^{-c/n_b}, \tag{7}$$

with ω a non-learnable buffer. The magnitude and the intrinsic phase ψ are whatever W_Q, W_K happen to produce; they are never constrained, and in particular $\tilde q$ may lie anywhere in $\mathbb{C}$. Substituting into (4) gives the familiar dependence on $s-t$ alone.

2.2 MapFormer: path integration replaces the clock

MapFormer keeps (2) and changes only what generates θ.

Step 1 — token to Lie-algebra increment. A low-rank map (rank $r = 2$; paper App. A.7, $W_\Delta = W_\Delta^{\text{out}} W_\Delta^{\text{in}}$) reads *every* token, action or observation:

$$\Delta_t \;=\; W_{\text{out}}\, W_{\text{in}}\, x_t \;\in\; \mathbb{R}^{H\times n_b}, \qquad W_{\text{in}} \in \mathbb{R}^{r\times d}, \quad W_{\text{out}} \in \mathbb{R}^{(Hn_b)\times r}. \tag{8}$$

Nothing tells the model which tokens are actions; it must *learn* $\Delta \approx 0$ on observations. The paper states only $r \ll d$, "for instance $r = 2$" (App. A.7).

> **Superseded.** An earlier note here recorded the bottleneck as load-bearing with $r = 1$ dropping the 2D task to 0.66, and marked it unsupported. It is still unsupported — no $r = 1$ run exists anywhere — but the rank axis is no longer untested: §2.5 reports an 8-seed sweep over $r \in \{2, 4, 8, 16, 32\}$ in which $r = 2$ is the *worst* setting. Read $r = 2$ below as the paper's prescription, not as a verified optimum.

Step 2 — integrate.

$$\boxed{\theta_t^{(h,c)} \;=\; \omega^{(h,c)} \sum_{u\le t} \Delta_u^{(h,c)} \;=\; \omega^{(h,c)}\ \mathrm{cumsum}(\Delta)_t^{(h,c)}.} \tag{9}$$

Because $SO(2)$ is abelian, the prefix product of rotations collapses to a *cumulative sum of angles*, which is what makes this an $O(\log T)$ parallel scan rather than a sequential recurrence:

$$\prod_{u\le t} R(\omega\Delta_u) \;=\; R\Big(\omega \textstyle\sum_{u\le t} \Delta_u\Big). \tag{10}$$

Commutativity is the entire computational argument, and it is also the limitation: a turn/forward action space, where displacement depends on accumulated heading, is *not* representable by (9).

Step 3 — frequency schedule. The ω are learnable, initialised geometrically over n_b blocks:

$$\omega_i \;=\; \omega_{\max} \left(\frac{1}{\Delta_{\max}}\right)^{\frac{i}{n_b-1}}, \qquad i = 0, \ldots, n_b - 1, \qquad \omega_{\max} = 2\pi, \quad \Delta_{\max} = \text{grid size}. \tag{11}$$

> **Correction to the published equation.** Eq. (11) differs from the paper (v1/v3 eq. 17, v4 eq. 18) in *two* places, both deliberate. First the **sign**: the paper prints a negative exponent, giving an *increasing* schedule that contradicts its own stated boundary condition $\omega_{\min} = \omega_{\max}/\Delta_{\max} \le \omega_i \le \omega_{\max}$. Second the **denominator**: the paper writes n_b with $1 \le i \le n_b$, so the endpoint $\omega_{\min}$ is never attained; eq. (11) uses $n_b - 1$ with $0 \le i \le n_b - 1$, which hits both endpoints exactly. The corrected form is what satisfies the paper's own inequality and what matches RoPE's $\theta_i = \text{base}^{-2i/d}$. Both defects survive into v4.

RoPE is a special case — of the functional form, not the encoding. If the model learns $\Delta_t \equiv 1$ for all t, (9) becomes $\theta_t = (t{+}1)\,\omega$ (the cumsum is inclusive; only differences enter (4), so the offset is immaterial). That is RoPE's *form*, with the token index as the clock. It is *not* RoPE identically, for three reasons: ω here is a trained parameter rather than a frozen buffer; its schedule spans $[2\pi/64,\, 2\pi] = [0.098,\, 6.28]$ against RoPE's $[10^{-4},\, 1]$, a ratio running from 6.3× at $c = 0$ to

$982\times$ at $c = n_b - 1$; and Δ_t is a rank-2 function of the embedding, so $\Delta \equiv 1$ is a hypothesis about what was learned, not a setting. The accurate statement is that path integration is **RoPE with a learned, content-dependent clock rate — which may be zero or negative — over a learned frequency spectrum**.

In terms of (5): MapFormer sets the **phase** and leaves the magnitude untouched.

2.3 Selective RoPE: the unconstrained generator

Selective RoPE (Movahedi et al., arXiv:2511.17388, ICLR 2026) was posted on 21 November 2025, three days before MapFormer, and neither paper cites the other in any version. Its angle is, in the present notation,

$$\theta_t^{(h,c)} \;=\; \tau \sum_{u \le t} \Big[\underbrace{\sigma(W_g x_u)}_{\text{gate}} \odot \underbrace{(\kappa * W_\omega x)_u}_{\text{smoothed increment}} \Big]^{(h,c)}, \tag{12}$$

with κ a causal depthwise kernel and τ a learned temperature. (We take the width to be 4; the paper states none, and §2.6 shows why that matters more than it looks.) It is (6) at $m = Hn_b$: 64 **channels integrating** 64 **independent signals**, against MapFormer's 64 readouts of one two-dimensional accumulator. Their *baseline* has no rank bottleneck, so no cross-channel constraint, and $K > 1$.

> **Their stability fixes converge on ours.** The paper reports that Selective RoPE trains unstably at higher learning rates and that this is remedied "by either placing a weight norm on the ω projection or using a low-rank MLP with rank 8 or 16". A documented variant of Selective RoPE is therefore *low-rank*, at a rank in the same range this project measures as sufficient. Calling it "the unconstrained generator" is accurate for the arm implemented here and for their headline design, and overstates the distance between the two papers.

> **Scope of every Selective-RoPE number below.** The arms here swap the *generator* and keep MapFormer's *placement* — the angle is computed once from the token embeddings before the blocks, not per head and per layer from that layer's queries. At one layer the distinction is nearly vacuous, since $q = W_Q\,\mathrm{LN}(x)$ is itself a learned linear map of the token. At depth it is not, and §6.3 records it as the one axis on which the two designs remain uncompared. *Correction:* an earlier version of this box also called the token-versus-query source a deviation. It is not — their own ablation moves to the input: "we find that this can be remedied by using the ℓ_2-normalised input x_t, instead of the queries q_t, to parametrise the rotation angles ω". Only the per-layer half of the placement gap is real. **A negative result here does not refute Selective RoPE**; it is evidence about the generator's components in this setting, on tasks their paper does not run.

2.4 The inclusion

The three generators are nested, and each inclusion *removes a constraint* rather than adding a mechanism:

$$\underbrace{\text{RoPE}}_{m=1,\ \varphi\equiv\text{const}} \quad\subset\quad \underbrace{\text{MapFormer}}_{m=r,\ \varphi\ \text{linear},\ K=1} \quad\subset\quad \underbrace{\text{Selective RoPE}}_{m=Hn_b,\ \varphi\ \text{gated},\ K>1}\ .$$

The right-hand inclusion is **exact**, and it is worth constructing rather than asserting. Given MapFormer's $(W_{\text{in}}, W_{\text{out}}, \omega)$, set Selective RoPE's parameters as follows.

1 $\kappa :=$ the delta at the *last* tap, $(0,0,0,1)$. The convolution is left-padded by $K-1$, so $\text{out}_t = \sum_j w_j\, u_{t+j-3}$ and this weight gives $\text{out}_t = u_t$: K **collapses** $4 \to 1$.

2 $W_g := 0$. The gate is then the *constant* $\sigma(b_g) \in (0,1)^{Hn_b}$ for *any* bias, and a constant is absorbed at step 3.

3 $W_\omega := \mathrm{diag}(\mathrm{vec}\,\omega)\, W_{\mathrm{out}} W_{\mathrm{in}} \,/\, \big(\tau\, \sigma(b_g)\big)$, row-wise. Legal because the full-rank parameterisation does not *constrain* rank, and weight normalisation represents any matrix (rows $g_i v_i / \|v_i\|$, with $g_i = 0$ for a zero row).

4 $b_\omega := 0$, or MapFormer acquires an index-RoPE component.

Verified against both implementations (`verify_inclusion.py`): $\max|\Delta \cos| = 1.5 \times 10^{-5}$ over a length-64 stream.

The reduction, line by line. Substituting the four settings into (12) and using that cumsum is linear:

$$
\begin{aligned}
\theta_t &= \tau \sum_{u \le t} \Big[\sigma(W_g x_u) \odot (\kappa * W_\omega x)_u\Big] && \text{Selective RoPE, (12)} \\
&= \tau \sum_{u \le t} \Big[\sigma(b_g) \odot (\kappa * W_\omega x)_u\Big] && \text{step 2: } W_g = 0 \\
&= \tau\, \sigma(b_g) \odot \sum_{u \le t} W_\omega x_u && \text{step 1: } \kappa = \delta\text{; constants leave the sum} \\
&= \tau\, \sigma(b_g) \odot \frac{\mathrm{diag}(\mathrm{vec}\,\omega) W_{\mathrm{out}} W_{\mathrm{in}}}{\tau\, \sigma(b_g)} \sum_{u \le t} x_u && \text{step 3} \\
&= \omega \odot W_{\mathrm{out}} W_{\mathrm{in}} \sum_{u \le t} x_u \;=\; \omega \odot \mathrm{cumsum}(\Delta)_t && \text{which is (9).}
\end{aligned}
\tag{13}
$$

Each line is an identity, not an approximation, which is what makes the inclusion exact rather than asymptotic.

> **Correction.** An earlier version of this paragraph called the gate step a *saturating limit*. It is not a limit: $W_g = 0$ makes the gate exactly constant for any bias, and the constant is absorbed. The inclusion is exact, not asymptotic.

The inclusion is on the function, not the state. Selective RoPE realises MapFormer's θ with a 64-dimensional state confined to an r-dimensional subspace: the constructed W_ω has rank 2, measured. This forces the definition in §2.5 to be taken on the **composite** token-to-phase map $A{\cdot}B$ where $\varphi(x) = Bx$, not on A alone — otherwise this very construction would read $\rho = Hn_b = 64$ while computing a rank-2 function.

The chain is not uniformly exact. RoPE sits inside Selective RoPE *unconditionally*: $W_\omega := 0$ with bias b gives $\theta_t = \tau(t{+}1)b$, an exact index clock (constant to 7×10^{-9}). It sits inside MapFormer only *conditionally*, because Δ has **no bias term** — so $\Delta_t \equiv \mathrm{const}$ requires the learned embeddings to share a projection onto W_{in}, which is a hypothesis about what training produced rather than a setting one can choose. The three further qualifications are in §3.

Consequently **MapFormer's generator is a constrained submodel of Selective RoPE's**, and every comparison between the two is a question about whether a constraint is worth keeping. It is never a question about expressivity, and no experiment here can be read as one.

2.5 Coherence rank, and the two thresholds

Define the **coherence rank** as the rank of the *composite* token-to-phase map, $\rho = \mathrm{rank}(A{\cdot}B)$ where $\varphi(x) = Bx$: the dimension of the subspace of $\mathbb{R}^{Hn_b}$ that the phase vector occupies as t varies. It is 1 for RoPE, r for MapFormer, and generically Hn_b for Selective RoPE. Taking the rank on A alone would be wrong — §2.4 constructs a Selective RoPE with $A = \tau I$ that computes a rank-2 function.

For MapFormer this is an identity rather than a reading. Because $\Delta_t = W_{\mathrm{out}} W_{\mathrm{in}} x_t$ and cumsum is linear,

$$\theta_t \;=\; \omega \odot \mathrm{cumsum}(W_{\mathrm{out}} W_{\mathrm{in}} x)_t \;=\; \omega \odot W_{\mathrm{out}} \underbrace{\sum_{u \le t} W_{\mathrm{in}} x_u}_{s_t \,\in\, \mathbb{R}^r}, \tag{14}$$

checked against the implementation to 9×10^{-7}. **All $Hn_b = 64$ phase channels are linear readouts of a single two-dimensional accumulator.**

Rank binds twice, for two unrelated reasons, and separating them is the point of this section.

(i) Geometry: a packing bound, with a correction

> **The first version of this argument was wrong, and its own gate falsified it.** It claimed that on a D-dimensional lattice the induced map $\mathbb{Z}^D \to \mathbb{R}^r$ has a kernel of dimension $D - r$, so that $r < D$ makes distinct positions *identical* in phase. That is false. The kernel of a generic linear map is a $(D - r)$-dimensional subspace of $\mathbb{R}^D$ containing no non-zero lattice point almost surely, so a generic W is injective on any finite set of positions and exact collisions never occur. The validator written to measure the collision rate returned 0.0% at every (D, r), which is what a correct gate on a wrong claim looks like. What actually binds is not identity but *separation*, below.

The phase enters attention only through $\cos(\theta_s - \theta_t + \psi)$, which is 2π-periodic in each block, so the code lives on the torus $\mathbb{T}^{Hn_b}$ — and by (6) it is confined to an *ρ-dimensional subtorus* of it. Packing the N^D positions of a D-torus into an ρ-dimensional one forces them close together: with the cloud normalised to unit scale, the minimum separation between distinct positions obeys

$$\boxed{\min_{p \ne p'} \big\| s(p) - s(p') \big\| \;\lesssim\; N^{-\max(D/\rho,\,1)}.} \tag{15}$$

Two regimes follow, and the exponent is the whole content of the bound.

- $\rho \ge D$: the accumulator can embed the position lattice faithfully, and the separation is set by the lattice's own minimal distance against a spread of order N. The exponent **saturates at** 1; more rank buys nothing geometric.
- $\rho < D$: the image is a projected lattice, dense in the subtorus, and the exponent is $D/\rho > 1$ — strictly worse, and worse the smaller ρ is.

Measured by projecting the full position set through random rank-ρ maps and fitting log(separation) against $\log N$:

D	ρ	D/ρ	predicted exponent	fitted
2	2	1.00	−1.00	−1.02
2	4	0.50	−1.00 *(saturated)*	−1.04
3	2	1.50	−1.50	−1.51
3	3	1.00	−1.00	−1.44
4	2	2.00	−2.00	−2.17
4	4	1.00	−1.00	−1.23

The $\rho < D$ rows track the prediction closely; the $\rho = D$ rows overshoot the floor, which is a finite-size effect — they are fitted over four map sizes with $N \leq 10$, where the asymptotic regime has not been reached. The saturation itself is confirmed by the D=2, ρ=4 row, whose exponent stays at -1 rather than falling to the naive -0.5.

> **What this does *not* explain.** At $D = 2$ both $\rho = 2$ and $\rho = 4$ sit at the floor exponent, so the bound predicts *no* advantage for $r = 4$ on a two-dimensional grid. One is measured, at $+0.085$ and 8/8 seeds. The geometric argument is therefore silent about the effect this project actually observed, which is the reason for (ii) — and a useful discipline, since a theory that explained the 2D gap as well would have been explaining it twice.

(ii) Optimisation: $\rho = D$ is expressible but not reachable

In 2D, $r = 2$ is exactly sufficient by (i) and is what the paper prescribes (r is the dimensionality of the action space, App. A.7). It nonetheless costs 0.085 accuracy at $T = 1024$ against $r = 4$ (8/8 seeds, $t = 3.57$), and the code it learns is skewed: opposite actions fail to cancel by half the action scale (0.495) and the two axes are close to parallel ($|\cos| = 0.783$), against 0.092 and 0.175 at $r = 4$. Yet the $r = 4$ latents still place 100.0% of their spectral energy in a 2-plane, and $r = 8, 16, 32$ are flat within noise of $r = 4$.

The extra rank carries no signal; it is slack that lets gradient descent reach an exact group homomorphism it otherwise misses. The representation is two-dimensional at convergence either way. The paper's structural argument for $r = 2$ is right about what is expressible and wrong about what is reachable, and the distinction is not rhetorical — it is the difference between a bound that saturates at $r = D$ and a measurement that does not.

Why coherence is cheap to keep

Nothing above says a *large* ρ is harmful in itself, and the direct comparison bears that out: the unconstrained generator's $\rho = 64$ buys no more on the torus than $r = 4$ does (§6), at 20× the parameters. The constraint is worth keeping because it is nearly free and because it is what makes the phase a multi-scale code for one quantity rather than 64 unrelated clocks — not because incoherence has been shown to cost. On parity, where the latent is a single bit and $\rho^* = 1$, the unconstrained generator does measurably worse; that is the only place in these experiments where incoherence has a price.

2.6 What $K > 1$ costs: the group structure

With $K = 1$ and φ linear, (6) is exactly the action of the free commutative monoid on tokens: the phase depends on the *multiset* of tokens traversed and on nothing else, so $\theta(a \text{ then } b) = \theta(b \text{ then } a) = \theta(a) + \theta(b)$, and $\prod_{u \leq t} R(\omega\Delta_u) = R(\omega \sum_{u \leq t} \Delta_u)$. Commutativity is MapFormer's entire computational argument, and it is also its stated limitation: a turn/forward action space, where displacement depends on accumulated heading, is not representable — measured, on MiniWorld, as -0.388 of a -0.438 swing attributable to rotation alone.

A convolution with $K > 1$ makes the increment of a token depend on its neighbours, so the phase ceases to be a function of the multiset and the map is no longer a homomorphism. The word "commutativity" does not appear once in Selective RoPE. The measured cost of giving it up is -0.020 on parity and -0.064 on the torus, for 193 parameters — the only knob in §6 that is free and negative on both tasks, and the only one whose sign was predicted.

The K axis is sharper than "blurring", and their own derivation says so. Selective RoPE motivates its convolution as a *shift*: "the shift over the queries q can be expressed by a 1d short-convolution", from an angle $\phi_j = \langle \omega_j,\, q_t - q_{t-1} \rangle$. That is a first difference, $K = 2$ with weights $(-1, +1)$ — and a first difference inside a cumulative sum *telescopes*:

$$\sum_{u \le t} \big(z_u - z_{u-1} \big) \;=\; z_t - z_0. \tag{16}$$

With the exact difference kernel the angle is a function of the *current* token alone and no accumulation is left. So K does not interpolate between sharp and blurred increments; it interpolates between **accumulation** ($K = 1$, identity kernel) and **no accumulation** (the differencer), and the kernel's DC gain says where a trained model sits.

> **Measured, and the answer is neither** (`CONV_KERNEL_PROBE.md`). Projecting the learned unit-normalised kernels onto the identity and onto the differencer gives 0.423–0.470 and 0.345–0.463 across both tasks, against 0.424 for a kernel drawn uniformly at random. The convolution does not learn to get out of the way, and does not learn the difference its own derivation motivates — it stays an arbitrary smear of the increment applied before accumulation, which is a sufficient account of the cost. **The width $K = 4$ used here is an assumption**, following the Mamba/GLA convention; the paper states no width in its pseudocode, its two ablation tables or its hyperparameter appendices, and $K = 2$ with a fixed difference kernel — the setting its derivation specifies — has not been run.

3 The magnitude axis: PoPE

PoPE (Polar Coordinate Positional Embeddings) makes (5) explicit rather than emergent.

> **Notation change, from here on.** §1 defined n_b as the number of *2D blocks*, $n_b = d_{\text{head}}/2$. PoPE indexes one frequency per *element*, so throughout this section and the next $n_b = d_{\text{head}}$. Read $\vartheta_c = \text{base}^{-c/n_b}$ below as $\text{base}^{-c/d_{\text{head}}}$; under §1's definition it would be wrong by a factor of two in the exponent.

Per element c:

$$\mu^q_{t,c} = \text{softplus}\big(Q_{t,c}\big), \qquad \mu^k_{s,c} = \text{softplus}\big(K_{s,c}\big), \qquad \text{(content, strictly positive)} \tag{17}$$

$$\varphi^q_{t,c} = t\,\vartheta_c, \qquad \varphi^k_{s,c} = s\,\vartheta_c + \delta_c, \qquad \text{(position, plus a bias)} \tag{18}$$

with $\vartheta_c = \text{base}^{-c/n_b}$ a **fixed buffer** and δ_c a learnable per-(head, frequency) constant.

> **δ_c is very nearly inert in practice.** It is clamped to $[-2\pi, 0]$ in the *forward pass*, not projected onto that range, so its gradient is identically zero outside the interval — and it is initialised at exactly 0, i.e. *on* the upper boundary. Any step that would push it positive kills its gradient permanently. Measured on trained checkpoints (4 heads $\times$ 64 frequencies $\times$ 4 layers), **91–98% of δ_c entries end at ≥ 0**, clamped to zero with dead gradient; per-layer means run -0.0003 to -0.018 rad. For roughly 95% of frequencies, (19) is effectively $\cos((s-t)\vartheta_c)$. Treat the bias as a nominal degree of freedom, not a working one.

The logit is

$$\boxed{a_{ts} \;=\; \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \text{softplus}(Q_{t,c})\ \text{softplus}(K_{s,c})\ \cos\big((s-t)\,\vartheta_c + \delta_c\big).} \tag{19}$$

Implemented via the angle-difference expansion, so no complex arithmetic is needed:

$$\cos^K_{s,c} = \cos\varphi_{s,c}\cos\delta_c - \sin\varphi_{s,c}\sin\delta_c, \qquad \sin^K_{s,c} = \sin\varphi_{s,c}\cos\delta_c + \cos\varphi_{s,c}\sin\delta_c, \tag{20}$$

$$\begin{aligned} a_{ts} = \tfrac{1}{\sqrt{d_{\text{head}}}} \Big[&(\mu^q \cos\varphi)_t \cdot (\mu^k \cos^K)_s^\top \\ &+ (\mu^q \sin\varphi)_t \cdot (\mu^k \sin^K)_s^\top \Big]. \end{aligned} \tag{21}$$

What PoPE does *not* do. Its phase is a function of position only. The magnitude is content-dependent; the *angle* is not, apart from the constant bias δ_c. This is worth stating because the name invites the opposite reading: PoPE is **not** in the CoPE / Selective-RoPE family, which learn content-dependent *angles*. It modifies the other factor.

In terms of (5): PoPE sets the **magnitude** and leaves the phase as a fixed clock.

3.1 The magnitude axis is also a scan: forget gates

Selective RoPE's own framing arrives at the two-factor decomposition from the opposite direction, and it is worth recording because it names a cell this note had left empty. Their stated design principle is *rotation and decay*: recall needs "(i) rotation, to encode relative position while preserving norms, and (ii) decay, to selectively discard past key–value associations", summarised as **"Real parts forget; imaginary parts encode position."** A complex diagonal state transition has a modulus and a phase; the modulus is a forget gate, the phase is a rotation.

That is (5)'s two factors, derived from diagonal SSMs rather than from the polar form of a rotary logit. But their magnitude is *accumulated* — a product $\prod_u \gamma_u$ over the transition — where PoPE's is *instantaneous*, a function of the current token only. Crossing that against the phase axis completes the design space:

	phase	**magnitude**
accumulated	MapFormer, Selective RoPE	forget gate / decay (GLA, Mamba, FoX)
instantaneous	ψ (free in RoPE; deleted by PoPE)	PoPE's softplus(Q)

MapFormer occupies one cell and has nothing in the accumulated-magnitude one. It uses softmax attention, which carries no decay, and adds no forget gate. On their reading that is a real omission rather than a stylistic difference — their classification of existing models runs the same way, with DeltaNet already possessing a rotational component and gaining from an added forget gate, and the softmax Transformer possessing rotation implicitly along random axes so that "only a forget gate is needed to fully satisfy the rotation + decay recipe", which is the Forgetting Transformer.

> **Which of their selectivities is MapFormer's.** *Selective rotation*, exactly: same factor, same accumulation, differing only in the (m, φ, K) settings of §2. *Selective decay* has no MapFormer counterpart at all. Whether it should is untested here — a forget gate on a navigation task with a bounded map is not obviously wanted, since discarding old key–value associations is the opposite of what a cognitive map is for.

Does anything on their side resemble MapEM? MapEM computes two attention matrices and multiplies them elementwise, $A = \mathrm{softmax}(A_X \odot A_P)$. Nothing in Selective RoPE does that, and the closest structural analogue is the forget gate rather than anything in the rotation. Writing their eq. (13) with a decayed transition, the weight linking query t to key τ factors as

$$\underbrace{\left(\prod_{u=\tau+1}^{t} \gamma_u\right)}_{\text{positional}} \times \underbrace{k_\tau^\top \check{R}_{\tau+1:t}\, q_t}_{\text{content} \times \text{position}}, \tag{22}$$

which has the *shape* of $A_X \odot A_P$ — a positional scalar gating a content term, an AND rather than an OR. Three things make it a weak analogue. The positional factor is a *monotone* decay, so it can only downweight with distance and can never peak at a particular displacement, which is the thing a cognitive map needs. It is itself content-dependent, since γ reads the intervening tokens, so it is not a clean position channel. And it falls out of the recurrence rather than being a decision to factor attention. **The two-attention-matrix construction has no counterpart in that literature**, which is consistent with the reading that MapEM's AND-gate is the genuinely distinct piece of the MapFormer paper.

4 MapPoPE: the conjunction

Since MapFormer determines φ and PoPE determines μ, combining them requires no new mechanism — only substituting one into the other. Replace PoPE's positional phase $t\vartheta_c$ with the path-integrated angle (9):

$$\boxed{a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \underbrace{\text{softplus}(Q_{t,c})\ \text{softplus}(K_{s,c})}_{\text{PoPE: magnitude from content}} \cos\Big(\underbrace{\theta_s^{(c)} - \theta_t^{(c)}}_{\text{MapFormer: path-integrated phase}} + \underbrace{\delta_c}_{\text{PoPE bias}} \Big),} \tag{23}$$

with $\theta_t^{(c)} = \omega^{(c)} \sum_{u \le t} \Delta_u^{(c)}$ as in (9). In code the class is three lines: inherit MapFormer's forward pass (so $\cos\theta, \sin\theta$ still come from the path integrator), widen the angle machinery, and swap the attention module.

The one non-trivial detail: frequency count. RoPE rotates *pairs* of coordinates, so it needs $n_b = d_{\text{head}}/2$ angles. PoPE assigns one frequency per *element*, so it needs $n_b = d_{\text{head}}$. The combination therefore rebuilds both position modules at the wider count:

$$n_b \ :\ d_{\text{head}}/2 \ \longmapsto\ d_{\text{head}}, \qquad W_{\text{out}} \in \mathbb{R}^{(H d_{\text{head}}) \times r}, \qquad \omega \in \mathbb{R}^{H \times d_{\text{head}}}. \tag{24}$$

In parameters this is small: at $d_{\text{model}} = 128$, $H = 2$, one layer, the two models are **204,373** and **204,693**, a difference of +320 (+0.16%) — +64 on ω, +128 on W_{out}, +128 for δ_c. The consequential change is not the parameter count but the **frequency count**: MapPoPE carries twice as many independent position frequencies, indexed per element rather than per pair. So the two are neither exactly parameter-matched nor frequency-matched, and the second is what a comparison has to contend with.

Why the parameter count barely moves: the content budget is conserved. The widening is not a capacity increase, and the arithmetic says why. Ordinary rotary attention groups d_{head} coordinates into $d_{\text{head}}/2$ complex numbers, each carrying a magnitude *and* an intrinsic phase — d_{head} real degrees of freedom of content per head. PoPE assigns one frequency to each of the d_{head} elements and keeps only a magnitude — again d_{head}. **PoPE spends the same content budget differently**: it buys twice the frequency resolution by giving up the per-block phase ψ. That is the same trade as the "restriction" noted in §3, counted rather than described, and it is the reason the two models differ by 320 parameters rather than by a factor.

> **What this does and does not contaminate.** In the 2×2 {RoPE, PoPE} $\times$ {index, path-integrated}, both PoPE cells are wide and both RoPE cells narrow, so the width confound is *constant along the encoding factor.* The position main effect and the interaction are therefore clean; only the encoding main effect carries it — and that is the small one (+0.011 on the torus against +0.461 for position). There is also a direct empirical check: if the extra frequencies were doing the work, PoPE with an index phase should beat RoPE with an index phase. It does not (0.509 vs 0.530, both on the 0.506 floor). The missing control is a *narrow* PoPE at $n_b = d_{\text{head}}/2$ with each angle repeated across its pair, which would isolate the parameterisation from the width; it has not been run.

The widening is a property of PoPE itself, not of the combination — the index-phase PoPE baseline also runs at $n_b = d_{\text{head}}$.

4.1 Why the two compose, and when they do not

(23) predicts the observed pattern.

- **PoPE alone is inert where position carries the signal.** On the torus navigation task, PoPE with an index phase scores 0.509 against a measured floor of 0.506. A better magnitude parameterisation is worth nothing when the phase encodes no spatial information.
- **PoPE plus path integration is the strongest configuration measured.** 0.994 on the same task. Once the phase carries genuine position, sharpening the content channel pays.
- **On text the picture does not survive its own metric.** Reading the *final* checkpoint gives PoPE alone −0.0053 bpc, path integration alone −0.0041, sum −0.0093, both together −0.0058 — apparently sub-additive. But the within-run checkpoint standard deviation is 0.003–0.007, larger than every one of those quantities, which is why the source file specifies the **mean of the last five checkpoints** instead. On that metric the numbers are −0.0041, −0.0006, sum −0.0047, both −0.0052: path integration alone nearly vanishes and the combination is mildly *super*-additive — the opposite sign. Both single-axis arms are also $n = 1$. **Nothing about additivity on text is established here in either direction**; the two readings are reported so the fragility is visible rather than resolved by choosing one.

The general statement: the two mechanisms are independent as *parameterisations* — they touch disjoint factors of (5) — but that does not make their *effects* independent. Whether they add depends on whether the task has separate work for a phase and a magnitude to do.

5 What leaves the frame

Two kinds of mechanism are outside (5), and it is worth being explicit about which, since both are routinely grouped with the ones above.

CoPE, and why it is not in this family. Contextual Position Encoding computes a gate from a *query–key pair*, $g_{ts} = \sigma(q_t \cdot k_s)$, and takes the position of s relative to t to be $\sum_{s<i\leq t} g_{ti}$. That quantity depends on both indices and on the query, so it cannot be written as $\theta_s - \theta_t$ for any per-token θ: the factorisation that (4) relies on — and with it the relative-position property that

holds for *any* θ — does not exist. **CoPE is not a phase generator with a different** (m, φ, K)**; it leaves the frame.** This is worth stating because CoPE, Selective RoPE and MapFormer are often listed together as "content-dependent position", and only the last two are the same machine.

PoPE is not in the CoPE family either, despite the name. Its magnitude is content-dependent; its *angle* is not, apart from the constant bias δ_c (§3). It modifies the other factor.

Mechanisms that act elsewhere in the network. Weight-shared recursion, hierarchical pooling and the InEKF correction are not choices about μ or φ at all: the first two change how the residual stream is processed at fixed θ, and the third post-processes θ before the rotation. Appendix A gives all three in this notation, together with what each was measured to be worth — which in the InEKF's case is a withdrawal.

6 What is measured

6.1 What the constraints are worth

One batch, 8 arms, the generator swapped and nothing else changed. Increments are against MapFormer at $r = 2$; MDE is $2.8\,\mathrm{sd}/\sqrt{n}$.

constraint removed	params	parity L=128	torus T=1024
$K = 1 \to 4$ (causal conv)	+193	−0.020 detectable	−0.064 (t −1.99)
$m = r \to Hn_b$ (full rank)	+7,873	−0.020 detectable	+0.058 (7/8)
φ linear → gated	+8,193	−0.030 detectable	+0.086 (8/8, sign p=0.008)
all three (Selective RoPE)	+16,385	−0.009 unmeasured	+0.048 (t 1.36)
$r = 2 \to 4$, *constraint kept*	+**384**	—	+**0.085** (8/8, t 3.57)

Three readings, in decreasing order of how well they are supported.

Rank is the only axis with a measurable effect, and it saturates at 4. $r = 4, 8, 16, 32$ are flat within noise (+0.085 to +0.095), so this is a step at $r = 2$, not a capacity slope. The same gain is available inside MapFormer's parameterisation for 21× fewer parameters than outside it.

The gate is not doing what its name suggests. It is statistically indistinguishable from the unrelated full-rank addition of comparable size (+0.018 at T=512 against an MDE of 0.026), so the parameters and not the mechanism are the plausible cause. The natural mechanism — that a sigmoid suppresses observation tokens, which MapFormer must instead learn to send to $\Delta \approx 0$ — was tested directly and falsified: the gate is 1.35× larger on actions than observations on the torus *where it helps*, and 1.54× larger on the analogous split on parity *where it hurts*.

The convolution is the one knob with a principled sign, and it is negative on both tasks at negligible cost. With $K > 1$ the increment of a token depends on its neighbours, so the phase ceases to be a function of the multiset of tokens traversed and the map from action sequences to $SO(2)^{n_b}$ is no longer a group homomorphism. Commutativity is MapFormer's entire computational argument — and the word does not appear once in Selective RoPE. The measured cost of giving it up is −0.020 on parity and −0.064 on the torus for 193 parameters.

6.2 A prediction, and the paper's own 5D negative

MapFormer v4 Table 6 reports MapWM collapsing in 5D: 0.75/0.50/0.35 on IID/OOD-d/OOD-s, *below* CoPE at 0.94/0.80/0.69, while MapEM-s holds at 1.00/1.00/0.87. The paper reports this without an account of it.

Equation (15) supplies a candidate. If $r = 2$ was held fixed at $D = 5$ — the paper specifies only $r \ll d$, "for instance $r = 2$" — then the phase code is confined to a 2-torus while the task needs a 5-torus, the separation exponent is $D/r = 2.5$ rather than the floor value 1, and position becomes progressively unresolvable as the map grows. MapEM's survival is consistent on the same reading: its multiplicative $A_X \odot A_P$ lets the content channel disambiguate what a degenerate position channel cannot.

> **How strong this is, stated plainly.** It is weaker than the kernel claim it replaces, in the way that a correct argument is often weaker than a wrong one. (15) predicts *degradation growing in D/r and in N*, not an impossibility, so it cannot say on its own whether 5D falls to 0.35 or merely slips. And their r in 5D is not observable from the paper. The strongest available claim after the experiment below is "$r = 2$ suffices to produce a 5D collapse of the reported shape in our reproduction", which is a candidate explanation for their number and not a demonstration of it.

Design. A D-dimensional torus (`environment_nd.py`): $2D$ actions $\pm e_i$, allocentric aliased observations, the interleaved stream and revisit-masked loss of the paper's task, and — mirrored exactly, because it sets the revisit statistics every 2D number here rests on — the *directed* walk, which draws an action and repeats it $k \sim U\{1..10\}$ times. State-space size is held near-constant so that only dimensionality moves: $D = 2$ at grid 32 (1024 cells), $D = 3$ at grid 10 (1000), $D = 5$ at grid 4 (1024). Arms $r \in \{2, D, D{+}2\}$, 8 seeds, one batch, the recipe of the 2D rank sweep.

What the two mechanisms predict, separately. The scale-free minimum separation at each cell of the design, from the gate:

	$D = 2$, N=32	$D = 3$, N=10	$D = 5$, N=4
$r = 2$	0.047	0.033	0.020
$r = D$	0.047	0.065	0.218
$r = D{+}2$	0.067	0.193	0.388
ratio r=D : r=2	1.0×	2.0×	**11**×

- **Geometry** predicts the r=2 deficit ordered D=5 $\gg$ D=3 > D=2, tracking the ratio row, and predicts *nothing* for $r = D$ versus $r = D{+}2$, which are both above the threshold.
- **Optimisation** predicts a further gap from $r = D$ to $r = D{+}2$ of roughly the size measured in 2D ($\approx$ 0.085 at 4× training length), present at every D and *not* growing with it.

The two are separable because they act on different cells of the same table, which is the reason for running all three ranks rather than two.

Falsifiers. $r = 2$ unimpaired at $D = 5$ kills the geometric account entirely and with it the explanation of Table 6. A flat r=D versus r=$D{+}2$ contrast at every D makes the optimisation half a 2D artifact. A deficit at $r = 2$ that does *not* grow with D leaves both halves standing but refutes (15) as the mechanism, since the exponent is the only thing D changes.

Gates, run before any GPU time (`ND_GATES.md`). All three configurations pass: an n-gram on the action stream alone at orders 1–5 reaches at most 0.536 against a measured majority-class rate of 0.526, and revisit rates are 0.23/0.28/0.62 with 30–79 scored positions per trajectory. The revisit gate is not a formality here: a simple random walk on $\mathbb{Z}^D$ is *transient* for $D \geq 3$ (Pólya), so revisits exist only because the torus is finite, and a configuration where they collapsed would show a "rank effect" that was really an absence of labels. The cross-D difference in revisit rate is a real confound and is the reason the primary contrast is *within* D; cross-D comparisons here are descriptive only.

6.3 The difference that has not been tested

Everything above concerns the *generator*. Selective RoPE also differs in *placement*: it derives the angle per head and per layer from that layer's own queries, whereas MapFormer computes θ once from the token embeddings and shares it across all layers and heads. The arms here swap the generator only, and at one layer the distinction is nearly vacuous, since $q = W_Q\,\mathrm{LN}(x)$ is itself a learned linear map of the token. At depth it is real, and it is the sole axis on which the two designs remain uncompared.

It is also not the axis probed by the refinement experiment of Appendix A: that carried a *bounded gated correction* to θ across iterations of a shared block and the optimiser declined to use it. Recomputing θ from scratch at each layer, from that layer's representation, is a different model and an open question.

6.4 Empirical anchors

mechanism	sets	parameter cost	measured effect
Path integration	phase θ	+448 on 199,042 (+0.23%)	+0.078 parity (16/16); +0.461 torus
PoPE	magnitude μ	$d_{\text{head}}/2 \rightarrow d_{\text{head}}$ widening, +320	+0.027 *with* path int. Without it, both arms are on the floor — see note
Bottleneck rank	state rank ρ	+384 on 204,373 (+0.19%)	+0.085 torus T=1024 for r=2 $\rightarrow$ 4 (8/8); flat to r=32
Recursion	block reuse	0	+0.056 parity (16/16); matches or beats 3× params
Hierarchy	token count	0 if weight-shared	length-dependent: ≈ 0 at $L = 16$, +0.060 at $L = 512$ (23/24 seeds); −22.8% time *at* $L = 2048$, but see note (iv)
InEKF	$\theta \rightarrow \hat{\theta}$	+49,600 on 204,373 (+24%)	no individual *component* is load-bearing; the whole filter is +0.06 to +0.13, at OOD length only

Reading the table. Parity figures are at $L = 128$ trained at $L = 16$, from one batch of 8 arms × 16 seeds. Torus figures are the position-axis factor average over both encodings (+0.4379 RoPE, +0.4851 PoPE). **Three qualifications the columns cannot carry.** (i) The hierarchy row is *not* a parity-at-$L = 128$ figure: its accuracy comes from runs trained at $L = 16$ and $L = 512$, and its time saving is measured at $L = 2048$ — at $L = 128$ the same benchmark shows +0.4%, i.e. no saving at all. (ii) The InEKF's parameter cost is measured on a 204,373-parameter torus model and the path-integration cost on a 199,042-parameter parity model; the two percentages are not commensurable, and the InEKF has never been run on parity. (iii) PoPE without path integration is a difference between two models sitting on the measured floor of 0.506 (0.509 vs 0.530), with a seed spread wider than the gap — it is a floor artifact, not an effect, and is omitted from the column for that reason. (iv) The −22.8% is measured on LoopedHourglass against the unshared flat scaffold, so it *bundles* the shared block's parameter saving with the pooling's compute saving.

Isolated on text, hierarchy alone is $1.23\times$ throughput ($\approx -18.7\%$) at -14.1% memory. Attributing -22.8% to the hierarchy axis alone overstates it.

A The other mechanisms, in the same notation

Weight-shared recursion (the "loop"). With B_ϑ a single transformer block and n the loop count,

$$h^{(0)} = \mathrm{Emb}(x), \qquad h^{(j+1)} = B_\vartheta\Big(h^{(j)};\, \cos\theta, \sin\theta\Big), \quad j = 0, \dots, n-1. \tag{25}$$

θ is computed *once* from the token embeddings and reused, matching both parents. Parameter count equals one layer; compute equals n layers.

Carrying a *bounded, gated correction* to θ across iterations — $\theta \leftarrow \theta_0 + g \cdot \tanh(\mathrm{refine}(h))$ with g initialised to zero — is a different model, and it is null: the learned gate settles at mean $|g| \approx 0.083$ with inconsistent sign across seeds, capping the correction at 0.14 rad against a θ spanning $\sim 2\pi T$. The gate was verified escapable (gradient 1.9×10^{-3} at zero), so the optimiser declined rather than failed to express it. A genuine per-iteration recomputation of $\omega \cdot \mathrm{cumsum}(\Delta(h^{(j)}))$ has *not* been run.

Level 1.5 invariant EKF (withdrawn; recorded for completeness). A correction applied to θ before the rotation:

$$R_t = \exp\big(\mathrm{clamp}(f_R(x_t), -5, 5)\big) \qquad \text{per-token measurement noise} \tag{26}$$

$$z_t = \pi \tanh\big(f_z(x_t)\big) \qquad \text{measurement in } [-\pi, \pi] \tag{27}$$

$$\nu_t = \mathrm{atan2}\big(\sin(z_t - \theta_t), \cos(z_t - \theta_t)\big) \qquad \text{wrapped innovation} \tag{28}$$

$$K_t = \Pi/(\Pi + R_t), \qquad \Pi = e^{\log \Pi} \qquad \text{gain; } \Pi \in \mathbb{R}^{H \times n_b}\text{, constant in } t \tag{29}$$

$$d_t = (1 - K_t)\, d_{t-1} + K_t\, \nu_t, \qquad d_{-1} = 0 \qquad \text{affine scan} \tag{30}$$

$$\hat{\theta}_t = \theta_t + d_t. \tag{31}$$

Π is a learned ($H \times n_b$) constant — 64 values at the default config, one per head and frequency block — not a scalar; it is constant across *tokens*, which is what distinguishes Level 1.5 from the full heteroscedastic filter. The recurrence is scalar in each channel and affine, so it is one more parallel scan.

Initialisation, which turned out to matter. $\log R$'s head is initialised with weights scaled by 0.01 and bias 0, giving $R \approx 1$ and hence $K \approx 0.5$ *at initialisation* — corrections at half strength before anything is learned. On the multiplicative-attention backbone this destroyed the gradient path and diverged 3 of 9 seeds; the fix was a bias of 3.0, i.e. $K \approx 0.05$, making the filter a near-no-op at init.

Status. No individual component is load-bearing: the measurement head, the per-token gate and the learned Π can each be removed alone at no measurable cost ($n = 5$). The whole filter does have an effect, but only at out-of-distribution length (+0.062 at T=512, +0.124 at T=1024, loss-matched), and on a task built to supply the drift it is meant to correct the effect *does not grow with the drift* — flat in one recipe, negative in another. The mechanism is withdrawn as inference; what survives is stabilisation.

Hierarchy (hourglass). With pooling factor k, a causal *shift-then-mean* pool P_k and an upsample U_k,

$$h \leftarrow B_{\text{pre}}(h), \qquad h \leftarrow h + U_k\, B_{\text{coarse}}\big(P_k h\big), \qquad h \leftarrow B_{\text{post}}(h), \tag{32}$$

with the coarse angle pooled alongside the content, and the sequence padded to a multiple of k then truncated after B_{post}.

Two details the notation would otherwise hide. P_k right-shifts by $k-1$ with a zero pad *before* averaging, so coarse token j is the mean over original indices $jk-1, \ldots, jk+k-2$ — not the plain group mean, and the first coarse token is dragged toward zero by the pad. And U_k is `repeat_interleave`, which is *not* the adjoint of P_k (that would carry a $1/k$ and the matching shift); it is written U_k rather than $P_k^{\dagger}$ for that reason.

Setting all three blocks to the *same* shared module gives the weight-shared hourglass: one block's parameters (204,373, identical to the flat 1-layer model, against 600,917 unshared) with the coarse block's token reduction retained. Note that every hourglass class ignores `n_layers` and is always this 3-block scaffold, so a flat control must be built at 3 blocks, not 1.